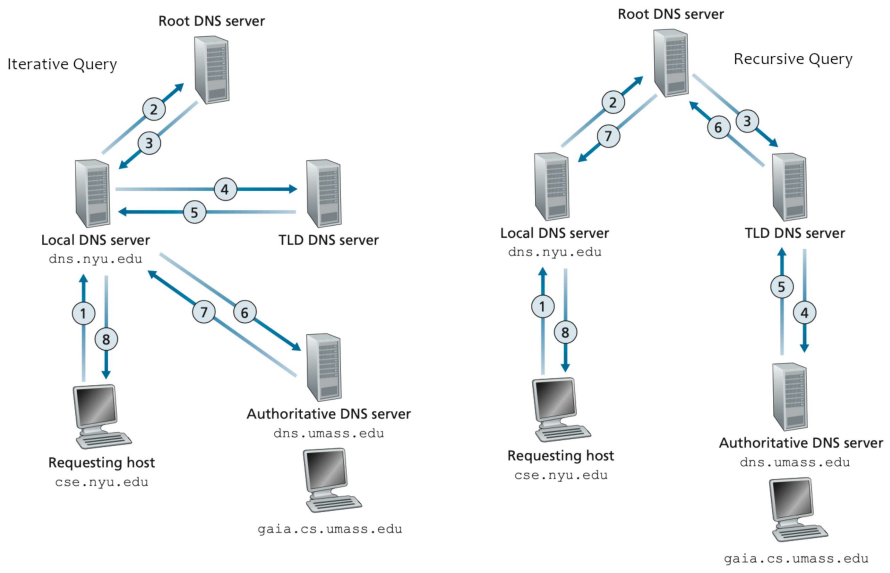
INTERACTIVE END-OF-CHAPTER EXERCISES

DNS - ITERATIVE VS RECURSIVE QUERY

Assume that a user is trying to visit gaia.cs.umass.edu, but his browser doesn't know the IP address of the website. In this example, examine the difference between an iterative and recursive DNS query.



QUESTION LIST

1. Between steps 1 and 2, where does the Local DNS server check first? Answer with 'User', 'DNS Local', 'DNS Root', 'DNS TLD', or 'DNS Authoritative'.

2. Between steps 2 and 3, assuming the root DNS server doesn't have the IP we want, where does the response link? Answer with 'DNS Local', 'DNS Root', 'DNS TLD', or 'DNS Authoritative'.

3. Between steps 4 and 5, assuming the TLD DNS server doesn't have the IP we want, where does the response link? Answer with 'DNS Local', 'DNS Root', 'DNS TLD', or 'DNS Authoritative'.

4. Between steps 6 and 7, the authoritative DNS server responds with the IP we want. What type of DNS record is returned?

5. Which type of query is considered best practice: iterative or recursive?

SOLUTION

1. The Local DNS server first checks the DNS Root.

2. The Local DNS server then checks the DNS TLD server.

3. Finally, the Local DNS server checks the DNS Authoritative server.

4. The DNS record received is type A (Type A is hostname:IP)

5. Iterative is considered 'best practice' because it puts less strain on the Root and TLD DNS servers.

QUESTION 1 OF 5

Between steps 1 and 2, where does the Local DNS server check first? Answer with 'User', 'DNS Local', 'DNS Root', 'DNS TLD', or 'DNS Authoritative'.

Answer

CHECK HINT SKIP

CHANGE QUERY TYPE

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